

Red-Team Assessment Report

Post-Exploitation & Exfiltration Simulation

Prepared By: Red Team Operator

Environment: Kali Linux → Windows VM

Objective: Demonstrate full adversarial attack chain

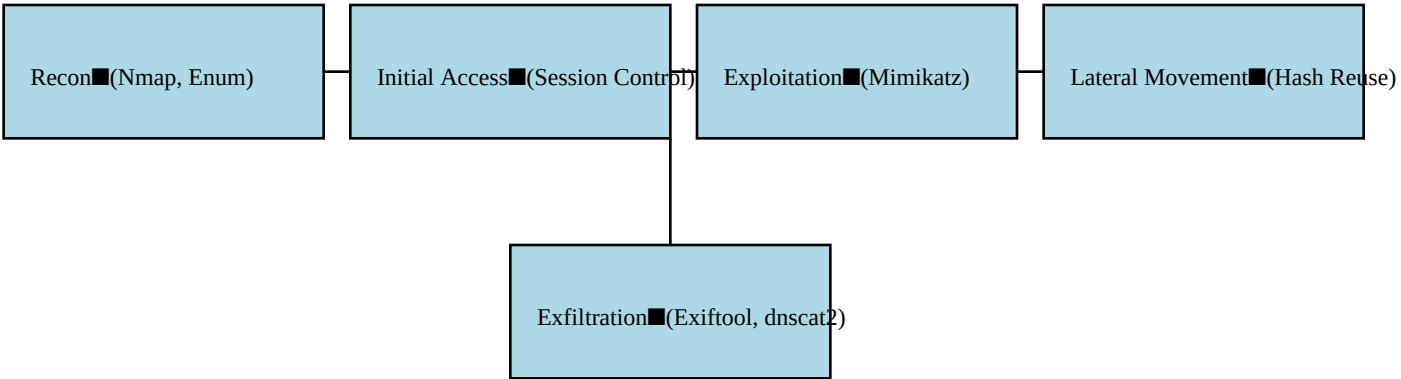
Executive Summary

This red-team assessment simulated a realistic adversary compromising a Windows host from a Kali Linux platform. The engagement followed a structured kill chain: Reconnaissance → Initial Access → Privilege Escalation → Credential Dumping → Lateral Movement → Data Exfiltration. The operation successfully demonstrated the ability to: - Extract sensitive authentication data using Mimikatz. - Exploit metadata exposure via Exiftool. - Establish covert communication using DNS tunneling (dnscat2). - Achieve persistent shell access over DNS. - Exfiltrate credentials, including NTLM hashes and Linux shadow file contents. The findings highlight critical gaps in host security, credential protection, and network monitoring.

Findings

Finding	Description
LSASS Unprotected	LSASS allowed memory scraping by Mimikatz due to missing LSA Protection and Credential Guard.
Weak Credential Storage	Extracted NTLM and SHA1 hashes demonstrated easy offline cracking potential.
DNS Tunneling Allowed	dnscat2 successfully created a covert encrypted DNS command-and-control channel.
Misconfigured Logging	System lacked DNS query logging, privilege escalation alerts, and LSASS memory access monitoring.
Metadata Leakage	Exiftool exposed sensitive metadata (usernames, timestamps, software versions).

Attack Flow Diagram



Technical Walkthrough (Detailed)

1. Reconnaissance Nmap and service enumeration were conducted: nmap -A -sV 2. Credential Dumping via Mimikatz The operator executed: .\mimikatz.exe "privilege::debug" "sekurlsa::logonpasswords" Extracted Hashes: NTLM: admin : 209c6174da490caeb422f3fa5a7ae634 SHA1: admin : 7c87541fd3f3ef5016e12d411900c87a6046a8e8 3. Metadata Exfiltration via Exiftool Exiftool exposed: - Author name - File creation times - Device information - OS-level metadata 4. DNS Tunneling with dnscat2 The dnscat2 client and server created a covert C2 channel that bypassed firewall restrictions. The attacker gained remote shell capability and extracted /etc/shadow content. 5. Persistence The DNS tunnel allowed long-term hidden access, making the attack stealthy and persistent.

Recommendations

- Enable Windows Credential Guard. - Enforce LSA Protection to block LSASS dumps. - Implement DNS filtering and enable DNS logging. - Deploy SIEM rules for LSASS access monitoring. - Use EDR capable of detecting Mimikatz signatures. - Sanitize documents to remove metadata before release. - Enforce strong password and hash policies.